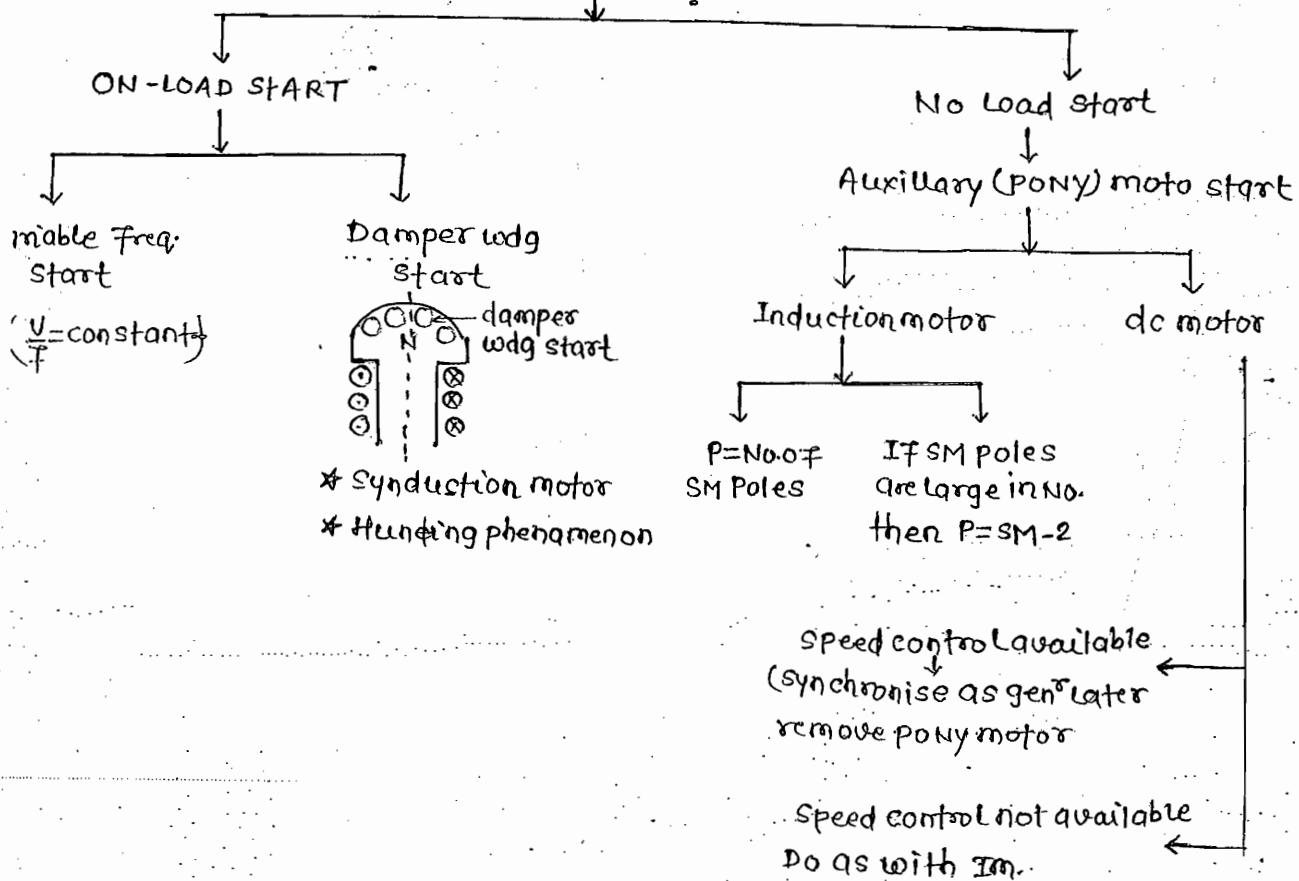


ATE-17/11/14

Synchronous motor starting



Que → A 3- ϕ Δ connected load takes 50A current at 0.707 lagging PF with 220V b/n the lines. A 3 ϕ Δ connected round motor syn. motor having a syn. reactance of $1.27\Omega/\phi$ is connected in parallel with the load. The power developed by motor is 33kW at a power angle of 30° . Calculate the overall PF of the motor & the load.

Soln →

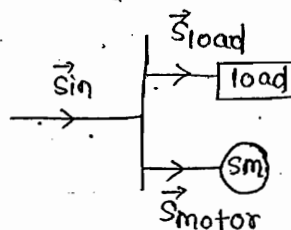
$$\vec{S}_{\text{load}} = \sqrt{3} \times 220 \times 50 / \cos^{-1}(0.707) \text{ VA}$$

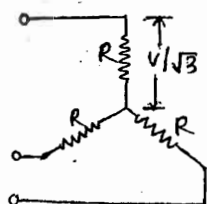
$$= 19.053 / 45^\circ \text{ kVA}$$

$$P = \frac{V \times E_f}{X_s} \sin \delta$$

$$330 \times 10^3 = \frac{220 \times E_f}{1.27} \sin 30^\circ$$

$$E_f = 381 \text{ volts (L-L)}$$





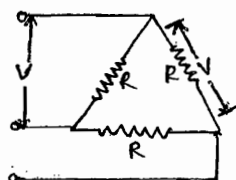
$$P_{1-\phi} = \frac{(V/\sqrt{3})^2}{R} = \frac{V^2}{3R}$$

$$P_{3-\phi} = 3 \times P_{1-\phi} = 3 \times \frac{V^2}{3R}$$

$$P_{3-\phi} = \frac{V^2}{R}$$

$$\boxed{P_{1-\phi} = \frac{V^2}{3R}}$$

$$\boxed{P_{3-\phi} = \frac{V^2}{R}}$$



$$P_{1-\phi} = \frac{V^2}{R}$$

$$P_{3-\phi} = 3 \times P_{1-\phi}$$

$$P_{3-\phi} = \frac{3V^2}{R}$$

$$\boxed{P_{1-\phi} = \frac{V^2}{R}}$$

$$\boxed{P_{3-\phi} = \frac{3V^2}{R}}$$

$$Q_{\text{motor}} = \frac{V}{X_s} (V - E_f \cos \delta)$$

$$= \frac{220}{1.27} (220 - 381 \cos 30^\circ)$$

$$= -19.047 \text{ kVAR (leading VARs)}$$

$$\vec{S}_{\text{motor}} = (33 - j19.047)$$

$$\vec{S}_T = \vec{S}_{\text{load}} + \vec{S}_{\text{motor}}$$

$$= 46.806 \angle -6.84^\circ \text{ kVA}$$

$$\text{Overall PF} = \cos(6.84) \text{ (leading)}$$

$$= 0.9929 \text{ (leading)}$$

$$\boxed{\text{PF} = 0.9929}$$

5.5
590

A 230V, 4P 50Hz Δ connected syn. motor has $R_a + jX_s = (0.6 + j3) \Omega/\phi$. Its field current is so adjusted that the motor draws 10A at UPF from rated vol. Now with field current unchanged load on the motor increase till it draws 40A from supply. Find the torque dev. & New PF.

Soln At UPF $E_f = \frac{230 \angle 0^\circ}{\sqrt{3}} - 10 \angle 0^\circ \times (0.6 + j3)$

$$= 130.29 \angle -13.81^\circ \text{ volt (L-N)}$$

When load is increased $\rightarrow$

$$130.29 \angle -\delta_2 = \frac{230 \angle 0^\circ}{\sqrt{3}} - 40 \angle -\phi_2 \times 3.06 \angle 78.89^\circ$$

$$130.29 \angle -\delta_2 = 132.79 - 122.4 \angle 78.69^\circ - \phi_2$$

$$= [132.79 - 122.4 \cos(78.69^\circ - \phi_2)] - j[122.4 \sin(78.69^\circ - \phi_2)]$$

Squaring & equating magnitudes

$$\cos(78.69^\circ - \phi_2) = 0.4811$$

$$\phi_2 = 17.45^\circ \text{ lagging}$$

$$\text{New PF} = \cos(17.45^\circ) \text{ lagging}$$

$$\boxed{\text{New PF} = 0.954 \text{ lagging}}$$

$$P_{dev} = P_{in} - \text{Total arm. cu. loss}$$

$$= \sqrt{3} \times 230 \times 40 \times 0.954 - 3 \times 40^2 \times (R_a = 0.6)$$

$$\boxed{P_{dev} = 12321.86 \text{ Watts}}$$

$$T_{dev} = \frac{P_{dev}}{\omega_{sm}}$$

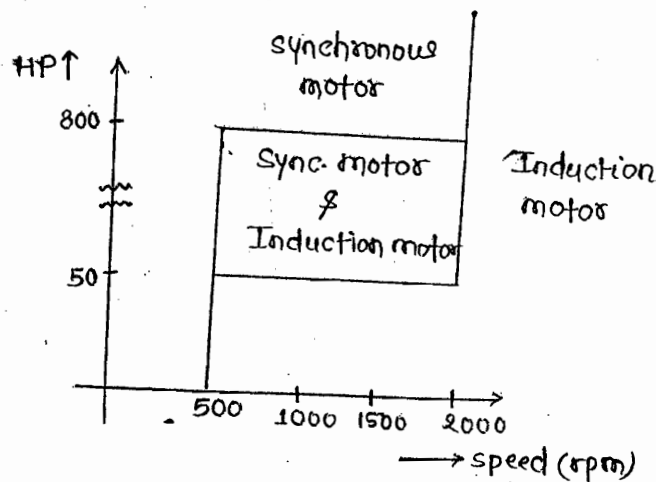
$$= \frac{12321.86}{\frac{2}{P} \times 2\pi f}$$

$$= \frac{12321.86}{(42) \times 2\pi \times 50}$$

$$\boxed{T_{dev} = 78.44 \text{ N-m}}$$

$$\left\{ \begin{aligned} \therefore \omega_{sm} &= \frac{2\pi N_s}{60} = \frac{2\pi}{60} \times \frac{1200}{P} \\ \omega_{sm} &= \frac{P}{P} \times 2\pi f \end{aligned} \right.$$

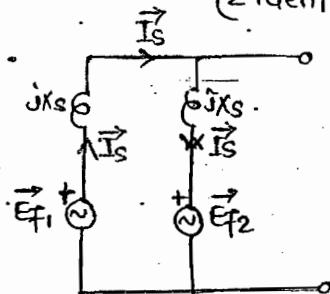
MOTOR APPLICATION GUIDE



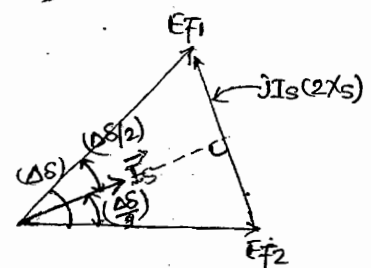
- * Below 500 rpm indⁿ motor not is use (12p) & above 2000 rpm syn. motor is not in use.
- * Below 50 HP Use Indⁿ motor & above HP 800 only syn. motor.
- * (50HP-800HP) & (500-2000) RPM both is available.
- * Squarmal cage SM is popular because of
- * SM speed is not effected by voltage where as IM highly affected by Vol. variation.

SYNCHRONISING POWER

(2 identical m/e on No-load)



$E_{f1} = E_{f2} = E_f$
Before disturbance



$$\sin \frac{\Delta \delta}{2} = \frac{I_s X_s}{E_f}$$

$$I_s = \frac{E_f}{X_s} \sin \frac{\Delta \delta}{2}$$

- * Due to incoming disturbance in the m/c there will be an excitation occurs & then E_{f2} will be excited by $(\Delta\delta)$ with E_{f1} .
- * m/c-1 (work as gen^r & source) will decelerate this increase in the excitation & the m/c-2 (work as motor & sink) will accelerate.
- * The amount of power which is going away from gen^r & coming into motor is equal in amount is known as synchronising power. i.e. the power used for synchronising the excitation, is &

$$\text{Synchronising power } P_s = \Delta P = E_f \cdot I_s \cdot \cos \frac{\Delta\delta}{2}$$

$$= E_f \left[\frac{E_f}{X_s} \sin \left(\frac{\Delta\delta}{2} \right) \right] \cos \left(\frac{\Delta\delta}{2} \right)$$

$$P_s = \Delta P = \frac{E_f^2}{2X_s} \sin(\Delta\delta) \text{ watts}$$

Since, $\Delta\delta$ is very small,

$$\sin(\Delta\delta) = \Delta\delta \text{ elect rad}$$

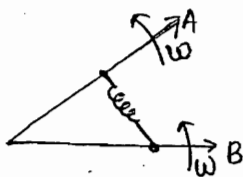
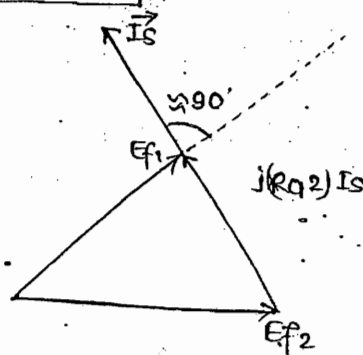
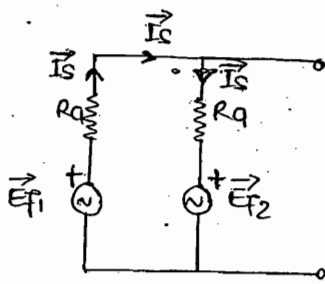
$$P_s = \Delta P = \frac{E_f^2}{2X_s} \times (\Delta\delta) \text{ watts}$$

Synchronising power coeff. $S_p = \frac{\Delta P}{\Delta\delta} = \frac{P_s}{\Delta\delta} \text{ watts/elect rad.}$

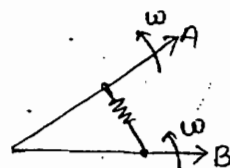
(OR)

Stiffness coeff

$$S_p = \frac{E_f^2}{2X_s} \text{ watts/elect rad.}$$

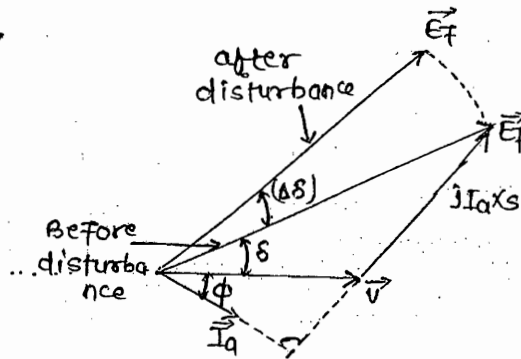
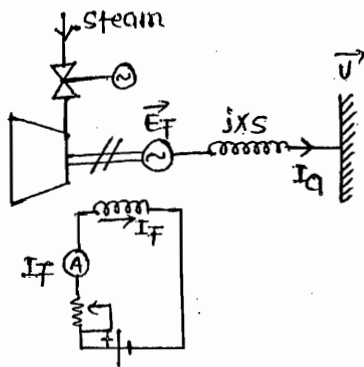


Restoring capability



No restoring capability.

* Synchronising Power $\rightarrow$ (Large m/c connected to ∞ bus)



Before disturbance; $P = \frac{VE_f}{X_s} \sin \delta$

After disturbance; $P + \Delta P = \frac{VE_f}{X_s} \sin(\delta + \Delta \delta)$

$\therefore$ synchronising power; $P_s = \Delta P = \frac{VE_f}{X_s} [\sin(\delta + \Delta \delta) - \sin \delta]$ watts

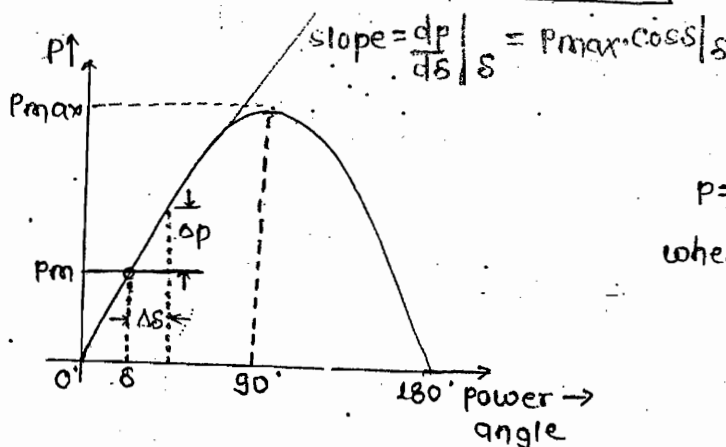
$$= \frac{VE_f}{X_s} [\sin \delta \cos \Delta \delta + \cos \delta \sin(\Delta \delta) - \sin \delta]$$

$$= \frac{VE_f}{X_s} [\cos \delta \sin(\Delta \delta) - \sin \delta (1 - \cos \Delta \delta)]$$

$$= \frac{VE_f}{X_s} [\cos \delta \sin \Delta \delta - \sin \delta \times 2 \sin^2 \frac{\Delta \delta}{2}]$$

since $\frac{\Delta \delta}{2}$ is very small; $\sin^2 \frac{\Delta \delta}{2} \approx 0$

$$P_s = \Delta P = \frac{VE_f}{X_s} \cdot \cos \delta \times \sin(\Delta \delta) \text{ watts}$$



$$P = P_{\max} \sin \delta$$

where $P_{\max} = \frac{VE_f}{X_s}$ at $\delta = 90^\circ$

Since δ is small, $\sin(\delta) = (\delta)$ elect rad

$$P_s = \Delta P = \frac{VE_f}{X_s} \cos \delta \times (\delta) \text{ watts}$$

$$\text{Hence } S_p = \frac{P_s}{\delta} = \frac{\Delta P}{\delta} \text{ watts/elect. Rad.}$$

$$S_p = \frac{VE_f}{X_s} \sin \delta$$

$$= P_{\max} \sin \delta$$

$$= \left. \frac{dP}{d\delta} \right|_{\delta = \text{initial operating angle}}$$

$$S_p = P_{\max} \text{ at } \delta = 0 \text{ (i.e. on NL)}$$

* This is the reason that it is recommended that the gen^r is synchronise On No Load.

* Captive power plant is a prisoner for own industries. i.e. generation of Power for only own industries.

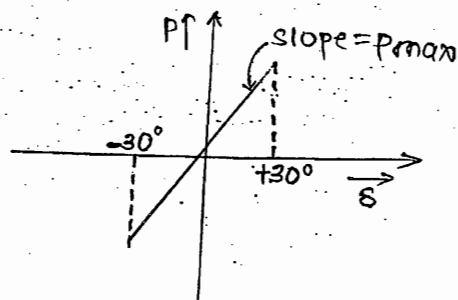
* Linearised Analysis →

$$P = P_{\max} \sin \delta$$

In the range $-30^\circ \leq \delta \leq +30^\circ$

$\sin \delta \approx \delta$ elect rad.

$$\text{Thus, } P = P_{\max} \times \delta \text{ watts}$$



* This is the reason that it is recommended that the syn. m/c be operated with power angle limited to 30° .

Que. → Cal. the synchronizing coefficient in kW & N-m/mech^o at FL for a 50 Hz 1000 KVA 0.8 PF lag, 6.6 KV BP & connected syn. gen^r of negligible resistance & syn. reactance of 0.8 pu.

Solⁿ →

$$\vec{E}_f = 1.0 + j(1.0) \angle -\cos^{-1}(0.8) \times 0.8$$

$$= 1.60 \angle 23.38^\circ \text{ pu} \approx 1.6125 \angle 23.39^\circ \text{ pu}$$

$$S_p = \left. \frac{E_f V}{X_s} \cos \delta \right|_{\delta = 23.38} = \frac{1.60 \times 1}{0.8} \cos(23.38)$$

$$= 1.836 \text{ pu}$$

$$\approx 1.85 \text{ pu per elect rad.}$$

$$S_p = 1.85 \times 1000 \text{ kw/elect rad.}$$

$$= 1850 \text{ kw/elect rad.}$$

$$= 1850 \times \frac{\pi}{180} \text{ kw/elect}^\circ$$

$$= 32.289 \text{ kw/elect}^\circ$$

$$= 129.156 \text{ kw/mech}^\circ$$

$$\text{Rad } \frac{\pi}{180} \text{ degree}$$

$$\text{Synchronising torque coeff.} = \frac{129.156 \times 10^3}{\frac{2}{3} \times 2\pi \times 50}$$

$$= 1644.465 \text{ Nm/mech}^\circ$$

$$F_{ar} = F_{ar(d)} + F_{ar(q)}$$

$$N_{ph} I_a = F_{ar(d)} + F_{ar(q)}$$

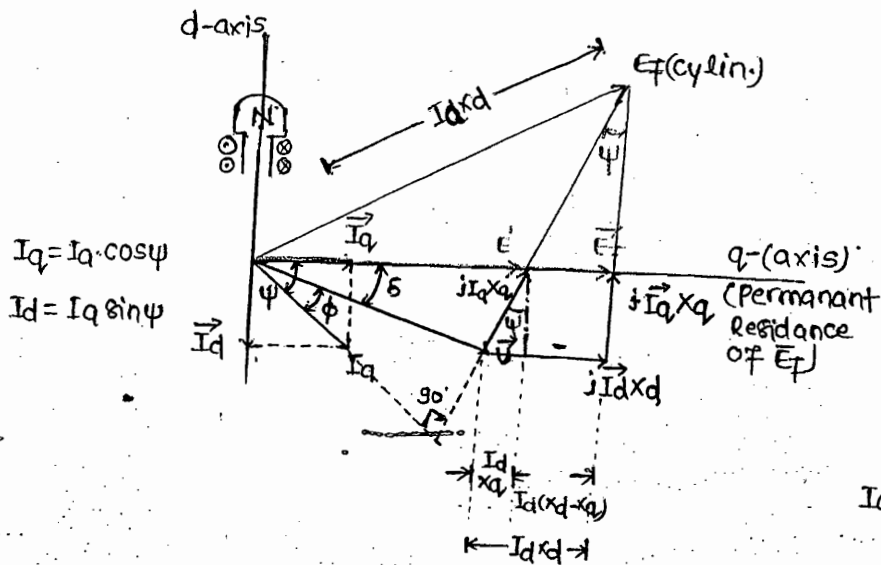
$$I_a = \frac{F_{ar(d)}}{N_{ph}} + \frac{F_{ar(q)}}{N_{ph}}$$

$$\vec{I}_a = \vec{I}_d + \vec{I}_q$$

$$x_q = 0.6 x_d \text{ to } 0.7 x_d$$

Salient Pole m/c

* Blondel's 2 reaction theory $\rightarrow$



The initial challenge.

Locating the q -axis

$$E' = V + jI_a X_q$$

$$\therefore E_f = E' + I_d(x_d - x_q)$$

(OR)

$$E_f = V \cos \delta + I_d X_d$$

$$I_q \times q = (I_q \times ?) \cos \psi$$

$$= (I_q \cos \psi) \cdot ?$$

$$= I q \times ?$$

$$? = xq$$

Over excited gen^r (PF lag).

$$P = VI \cos \phi$$

$$= VI \cos(\psi - \phi)$$

$$= V I_a [\cos \psi \cdot \cos \delta + \sin \psi \cdot \sin \delta]$$

$$= V \cos \delta (I_a \cos \psi) + V \sin \delta (I_a \sin \psi)$$

$$= V \cos \delta \cdot I_q + V \sin \delta I_d$$

$$= \frac{V \cos \delta}{x_q} \cdot I_q x_q + \frac{V \sin \delta}{x_d} \cdot I_d x_d$$

$$= \frac{V \cos \delta}{x_q} V \sin \delta + \frac{V \sin \delta}{x_d} (E_f - V \cos \delta)$$

$$= \frac{VE_f}{x_d} \sin \delta + V^2 \sin \delta \cos \delta \left(\frac{1}{x_q} - \frac{1}{x_d} \right)$$

$$= \frac{V \cdot E_f}{x_d} \sin \delta + \frac{V^2}{2} \sin 2\delta \left(\frac{1}{x_q} - \frac{1}{x_d} \right)$$

$$p = \frac{V \cdot E_f}{x_d} \cdot \sin \delta + \frac{V^2}{2} \sin 2\delta \left(\frac{1}{x_q} - \frac{1}{x_d} \right)$$

Excitation
Power

↓ Power due to Saliency (OR)

Reluctance power